MINISTRY OF EDUCATION AND TRAINING
CAN THO UNIVERSITY
COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGY

GRADUATION THESIS
BACHELOR OF ENGINEERING IN INFORMATION TECHNOLOGY
(HIGH-QUALITY PROGRAM)

DEVELOPING A WEB APPLICATION FOR MUSIC GENERATION FROM VIDEO USING NATURAL LANGUAGE AS INTERMEDIARY USING OPEN SOURCE AI MODELS

Student: Phan Trung Thuan
Student ID: B2111957
Class: 2021-2025 (K47)
Advisor: Dr. Lam Nhut Khang

Can Tho, 12/2025

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TRƯỜNG ĐẠI HỌC CẦN THƠ
TRƯỜNG CÔNG NGHỆ THÔNG TIN
VÀ TRUYỀN THÔNG

CỘNG HÒA XÃ HỘI CHỦ NGHĨA VIỆT NAM
Độc lập – Tự do – Hạnh phúc

XÁC NHẬN CHỈNH SỬA LUẬN VĂN THEO YÊU CẦU CỦA HỘI ĐỒNG

Tên luận văn (tiếng Việt và tiếng Anh):

Phát triển ứng dụng web cho tác vụ sinh nhạc từ video sử dụng ngôn ngữ tự nhiên làm trung gian bằng các mô hình AI mã nguồn mở

Developing a web application for Music Generation from video using Natural Language as Intermediary by open source AI Models

Họ tên sinh viên: Phan Trung Thuận MASV: B2111957

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Luận văn đã được chỉnh sửa theo góp ý của Hội đồng.

Cần Thơ, ngày 27 tháng 12 năm 2025
Giáo viên hướng dẫn
(Ký và ghi họ tên)

Lâm Nhựt Khang

COMMENTS OF THE ADVISOR

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Can Tho, Dec ……. 2025

Advisor

Lâm Nhựt Khang

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ABSTRACT

This report presents a language-mediated framework for video-to-music generation, aiming to automatically generate background music that is semantically aligned with video content. Unlike conventional approaches that directly fuse visual and audio features, the proposed system employs natural language as an intermediate representation to bridge video understanding and music generation. The framework integrates scene segmentation, video captioning, music feature inference, and prompt-based music generation using open-source models, enabling improved interpretability and controllability.

Experimental evaluation is conducted on a subset of the SymMV dataset, with vocals removed to focus on background music. The system is assessed using both audio quality metrics and cross-modal video-music relationship metrics based on ImageBind embeddings. Results show that the proposed approach achieves strong global semantic alignment and consistent temporal correspondence between video and generated music, outperforming a state-of-the-art baseline in several semantic alignment metrics. Although beat-level synchronization remains limited, the generated music exhibits stable spectral characteristics suitable for background accompaniment. Overall, the results demonstrate that natural language serves as an effective intermediary modality for semantically coherent video-to-music generation in web-based and interactive applications.

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
LLM	Large Language Model
SLM	Small Language Model
VLM	Vision-Language Model
CNN	Convolutional Neural Network
RNN	Recurrent Neural Network
LSTM	Long Short-Term Memory
GAN	Generative Adversarial Network
FFT	Fast Fourier Transform
STFT	Short-Time Fourier Transform
THD	Total Harmonic Distortion
SF	Spectral Flatness
SCV	Spectral Centroid Variability
ZCR	Zero-Crossing Rate
CMEC	Cross-Modal Embedding Correlation
BMS	Beat-Motion Synchrony
CMSA	Cross-Modal Similarity Alignment
SSM	Self-Similarity Matrix
GPU	Graphics Processing Unit
JSON	JavaScript Object Notation
BPM	Beats Per Minute

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CHAPTER 1: INTRODUCTION

This chapter introduces the overall background and motivation of the thesis. It outlines the importance of music in audiovisual media and highlights the challenges of automatically generating background music for video content. The chapter then defines the research problem, clarifies the objectives of the study, and positions the proposed work within the context of existing video-to-music generation research. Finally, it provides an overview of related work to establish the foundation for the proposed language-mediated approach.

1.1 Problem of the study

Music is a fundamental component in audiovisual media, playing a crucial role in shaping narrative flow, emotional expression, and audience engagement. In films, advertisements, games, and social media content, background music is often used to reinforce mood, guide attention, and enhance storytelling. However, the process of composing or selecting suitable music for a given video remains largely manual, time-consuming, and highly dependent on subjective human judgment and musical expertise. This creates a significant barrier for content creators who lack professional music production skills or resources.

With the rapid advancement of generative artificial intelligence, recent research has demonstrated promising results in both text-to-music and video-to-music generation. Nevertheless, many existing systems either directly map visual features to audio signals or rely on fixed emotion labels, limiting interpretability and user control. In creative applications, users often wish to refine or adjust the generated music based on abstract concepts such as atmosphere, narrative intention, or stylistic preference-elements that are more naturally expressed through language rather than low-level numerical features.

The primary purpose of this study is to develop an open-source, end-to-end web-based system capable of automatically generating background music from video content by using natural language as an intermediate semantic representation. Instead of directly coupling video features with music generation, the proposed framework explicitly separates video understanding, semantic reasoning, and audio synthesis into modular stages. Video content is first analyzed and summarized into descriptive language, which is then transformed into high-level musical attributes and generation prompts.

By adopting natural language as a bridging modality, the system aims to achieve three key objectives. First, it enhances semantic alignment, ensuring that the generated music reflects not only visual motion but also narrative context and emotional tone. Second, it improves interpretability and controllability, allowing users to understand and optionally modify the

reasoning process behind music generation. Third, it promotes accessibility and reproducibility by relying entirely on open-source models that can be deployed in resource-constrained environments such as web applications.

Ultimately, this study seeks to demonstrate that language-mediated video-to-music generation is a practical and effective alternative to traditional end-to-end multimodal approaches, particularly for interactive and real-world multimedia applications.

1.2 The goal of the thesis

The goal of this thesis is to design and implement an open-source, end-to-end system for automatic background music generation from video content, using natural language as an intermediate semantic representation. The proposed system aims to bridge the gap between video understanding and music generation by explicitly modeling high-level semantics instead of relying solely on low-level visual or motion features. Specifically, this thesis pursues the following objectives:

First, to develop a language-mediated video-to-music generation pipeline that decomposes the task into interpretable stages, including scene segmentation, video captioning, music feature inference, and prompt-based music generation. By introducing natural language as an explicit intermediary, the system enables clearer semantic reasoning and modular design.

Second, to improve semantic and emotional alignment between video content and generated music. Rather than focusing only on beat-level synchronization, the system seeks to capture narrative context, scene atmosphere, and emotional progression, ensuring that the generated background music supports the storytelling intent of the video.

Third, to enhance interpretability and controllability for users. The proposed framework allows users to inspect and optionally modify intermediate textual representations such as video captions and music feature descriptions, enabling creative control and reducing the “black-box” nature of end-to-end generative models.

Fourth, to ensure practical deployability in real-world applications. The system is implemented as a web-based application and relies entirely on open-source models, balancing generation quality with computational efficiency to support interactive usage and reproducibility.

Finally, the thesis aims to experimentally evaluate the effectiveness of the proposed approach using objective music quality metrics and cross-modal video-music relationship metrics. Through quantitative analysis and comparison with a state-of-the-art baseline, the study seeks to demonstrate that

language-mediated video-to-music generation is a viable and effective alternative to traditional multimodal generation methods.

1.3 Related Work

Recent research on video-to-music generation has attracted increasing attention due to advances in multimodal representation learning and generative audio models. Existing approaches can be broadly categorized based on how visual information is mapped to musical content, ranging from direct low-level conditioning to high-level semantic reasoning frameworks.

Early video-to-music generation systems primarily focused on direct visual-to-audio mappings, where low-level visual cues such as motion intensity, scene transitions, or human body movements were used to condition music generation. Representative works include Foley Music by Gan et al. [1], which maps human pose keypoints extracted from dance videos to MIDI sequences using Graph Neural Networks and Transformers. Similarly, D2M-GAN [2] employs vector-quantized audio representations and adversarial training to generate music conditioned on motion features derived from dance videos. These methods demonstrate strong rhythmic correspondence between visual motion and musical beats, making them effective for motion-centric scenarios such as dance or sports videos. Another line of research emphasizes vision-to-music conversion across different modalities. Wang and his team [3] highlighted systems translating general videos, human movements, and images into either symbolic or audio music, underscoring the importance of emotional coherence.

However, motion-driven approaches are inherently domain-specific and struggle to generalize to videos where motion is not the dominant semantic signal. In cinematic storytelling, dialogue-heavy scenes, or emotionally driven narratives, low-level motion features often fail to capture narrative intent or emotional progression. As a result, music generated by these methods may exhibit acceptable rhythmic synchronization while remaining semantically shallow or emotionally misaligned with the video content.

To address these limitations, subsequent research shifted toward semantic alignment-based approaches. VMAS (Video-to-Music Alignment System) proposed by Lin et al. [4] represents a key milestone in this direction. VMAS introduces contrastive learning to align video and music embeddings in a shared latent space, ensuring that semantically related video-music pairs are closer than unrelated ones. An autoregressive Transformer is then employed to generate music conditioned on these aligned representations, enabling both beat-level synchronization and higher-level semantic correspondence. Complementary to this work, Ji et al. [5] provided a comprehensive survey that categorizes video-to-music systems by model architecture (RNNs, GANs,

Transformers, Diffusion models) and conditioning mechanisms, highlighting the growing importance of semantic understanding in this task.

Emotion-aware conditioning further extends semantic alignment by explicitly modeling affective states. Video2Music by Kang et al. [6] employs an affective multimodal Transformer that predicts music attributes such as note density and loudness from visual emotion cues. V2Meow [7] emphasizes synchronization with visual beats and emotional consistency, while FilmComposer [8] targets silent film scoring using a multi-agent framework driven by large language models (LLMs). a simple yet controllable transformer-based model for high-quality symbolic and audio outputs. Zuo introduced GVMGen [9], which applies hierarchical attention mechanisms to capture both spatial and temporal video features for conditional music generation. FilmComposer simulates professional music production workflows by generating textual scene descriptions, emotional analyses, and compositional plans before producing music, demonstrating the effectiveness of language-based reasoning for capturing narrative structure and musical intent.

Parallel to video-conditioned music generation, rapid progress in text-to-music generation has significantly influenced the field. Models such as MusicGen [10], MusicLM [11], and DiffRhythm [12] demonstrate that natural language can serve as a powerful and flexible control modality for music synthesis. Zhao [13] reviewed symbolic and audio-based methods, discussing future directions for integrating singing voice synthesis. These systems allow users to specify genre, mood, instrumentation, and style using free-form text, resulting in music that is both high-quality and semantically interpretable. Mao and his colledges [14] explored cross-modal contrastive learning for robust video-music embeddings. Diffusion-based approaches such as Diff-BGM [15] further improve audio fidelity and temporal smoothness by incorporating segment-aware cross-attention mechanisms. Despite their strong performance, diffusion-based video-to-music systems are often computationally expensive and operate as end-to-end black boxes, limiting their suitability for interactive or web-based applications.

Beyond early motion-driven systems, recent video-to-music generation research has increasingly emphasized semantic consistency, temporal alignment, and scalability. MuVi [16] proposes a semantic and temporal alignment framework that jointly models visual scene semantics and rhythmic structure to guide music generation. Instead of relying solely on low-level motion cues, MuVi [16] extracts scene-level representations and aligns them with music embeddings in a shared latent space, improving global coherence across long video sequences. However, the method remains end-to-end and

lacks interpretability in how specific semantic elements influence musical decisions. VidMuse [17] introduces a simplified yet scalable framework that decomposes video-to-music generation into long-term and short-term modeling components. By explicitly separating global narrative structure from local rhythmic dynamics, VidMuse [17] achieves stable generation over extended videos. Nevertheless, the framework does not explicitly model high-level concepts such as emotion progression or narrative intent, which limits its ability to handle story-driven or dialogue-centric videos.

VeM (Video Echoed in Music) [18] further explores semantic and rhythmic correspondence by jointly optimizing temporal alignment and semantic similarity objectives. The authors demonstrate that explicitly modeling beat-level synchronization alongside semantic alignment improves perceptual coherence. Despite these improvements, VeM [18] still relies on latent embedding alignment and does not provide a mechanism for reasoning about abstract video content, such as themes or symbolic meanings.

Diffusion-based approaches have also been explored for video-to-music generation. Diff-V2M [19] introduces a hierarchical conditional diffusion model that separately handles global musical structure and local acoustic details. This design improves audio fidelity and temporal smoothness but significantly increases computational cost. Moreover, diffusion-based methods typically operate as black-box generators, making them difficult to deploy in real-time or interactive systems. Complementary to video-based approaches, image-to-music generation via cross-modal contrastive learning demonstrates that even static visual semantics can inform musical generation. These works highlight the importance of semantic abstraction over raw visual signals, reinforcing the need for intermediate representations that can capture meaning beyond motion. Overall, while recent video-to-music models improve alignment and audio quality, most approaches remain end-to-end and latent-space driven, offering limited interpretability and controllability. This motivates alternative frameworks that explicitly model semantic reasoning and allow higher-level control over music generation

Another emerging research direction explores language models as intermediaries for music understanding and reasoning. Works such as ChatMusician [20] and MusicBERT [21] demonstrate that large language models can reason about symbolic music structures, music theory, and compositional intent using textual representations. These models bridge the gap between human-centric musical concepts and machine-level generation, enabling explainable and controllable music synthesis. However, most existing systems either focus on text-only inputs or rely on proprietary APIs, reducing reproducibility and accessibility.

Collectively, these studies reveal several key trends. First, there is a clear transition from low-level motion-based conditioning toward high-level semantic and narrative alignment. Second, natural language has emerged as a unifying modality that enables interpretability, controllability, and modular system design. Third, despite promising results, few existing works provide an open-source, end-to-end framework that integrates video understanding, language-based reasoning, and music generation in a computationally efficient and deployable manner.

This thesis builds upon these insights by proposing a language-mediated video-to-music generation pipeline, where natural language explicitly serves as an intermediate representation between video content and music synthesis. Unlike prior end-to-end or motion-centric approaches, the proposed system decouples semantic reasoning from audio generation, enabling improved alignment quality, user interpretability, and practical deployment in web-based applications using fully open-source models.

1.4 The research method

The research method of this thesis is mainly based on studying and synthesizing reference materials from the Internet and reputable Information Technology community websites. These materials include research papers, technical documents, and open-source project documentation related to video processing, natural language processing, and music generation.

Based on the collected references, the knowledge and techniques learned during undergraduate studies are applied to analyze and solve the problem discussed in this thesis. The research focuses on applying existing methods and tools rather than developing new algorithms, with an emphasis on system integration and practical implementation.

The proposed approach follows an applied and development-oriented methodology. Open-source models and frameworks are selected and combined to build an end-to-end system for generating background music from video content. Experimental implementation and testing are conducted to observe system behavior and improve semantic alignment between video and generated music.

Finally, the effectiveness of the proposed system is evaluated through experimental results and comparison with baseline methods. This research method ensures that the thesis demonstrates both theoretical understanding and practical applicability.

1.5 Outline of thesis

This thesis is organized into five chapters, structured in 5 chapters: **Chapter 1 - Introduction** presents the background and motivation of the study. It defines the research problem, states the objectives of the thesis, describes the research methodology, and positions the proposed work within the context of existing video-to-music generation research. **Chapter 2 - Literature Review** provides a comprehensive review of related work in generative AI for music, video-to-music generation, emotion-conditioned music synthesis, text-to-music models, singing voice generation, and music understanding using large language models. This chapter identifies key research gaps that motivate the proposed approach. **Chapter 3 - Proposed Approaches** describes the architecture and implementation of the proposed language-mediated video-to-music generation pipeline. It details each system component, including scene segmentation, video captioning, music feature proposal, prompt preparation, music generation, output rendering, and evaluation metrics. **Chapter 4 - Experiment Results** presents the experimental setup, evaluation dataset, and quantitative results. The chapter analyzes music quality, video-music alignment, and computational efficiency, and compares the proposed system with a state-of-the-art baseline. **Chapter 5 - Conclusion and Future Work** summarizes the main contributions of the thesis, discusses its limitations, and outlines potential directions for future research, including improved temporal synchronization, user interaction, and integration with more advanced generative models.

CHAPTER 2: LITERATURE REVIEW

This chapter presents a comprehensive review of previous studies related to music generation and video-to-music generation. It covers key research directions including generative AI for music, video-conditioned music generation, emotion-conditioned models, text-to-music systems, singing voice synthesis, and music understanding using large language models. Through this review, the chapter identifies limitations in existing approaches and highlights research gaps that motivate the proposed language-mediated framework.

2.1 Generative AI for Music

Generative Artificial Intelligence (AI) for music has evolved through several major paradigms, reflecting broader advances in machine learning, representation learning, and multimodal modeling. The primary goal of music generation systems is to automatically produce musically coherent content that exhibits meaningful structure, stylistic consistency, and perceptual plausibility. Over time, research has transitioned from rule-based composition systems to data-driven deep learning approaches capable of learning complex musical patterns directly from large-scale datasets.

Early computational music generation relied on symbolic representations, where music was encoded as sequences of discrete events such as notes, chords, durations, and velocities. These systems were often based on hand-crafted rules, grammars, or probabilistic models such as Markov chains. While rule-based approaches provided interpretability and theoretical grounding, they lacked scalability and struggled to capture expressive variation. The introduction of deep learning, particularly Recurrent Neural Networks (RNNs), marked a turning point by enabling models to learn temporal dependencies directly from data. Long Short-Term Memory (LSTM) networks, in particular, demonstrated strong performance in modeling sequential musical structures such as melody and harmony.

Representative symbolic-generation systems include DeepBach [22], which successfully models four-part Bach-style chorales by learning local harmonic constraints and global stylistic consistency. Similarly, early generative adversarial approaches such as MuseGAN [23] introduced multi-track symbolic generation, allowing simultaneous synthesis of drums, bass, melody, and accompaniment. These models demonstrated that neural networks could learn multi-voice coordination and rhythmic structure. However, RNN-based and GAN-based symbolic models often suffered from limited long-term coherence, difficulty in controlling global form, and sensitivity to sequence length, which constrained their applicability to longer compositions.

To address these limitations, Transformer architectures were introduced into music generation. Transformers replace recurrent connections with self-attention mechanisms, allowing models to capture long-range dependencies more effectively. This architectural shift proved particularly beneficial for music, where motifs, harmonic progressions, and thematic structures often span long temporal intervals. Systems such as MuseNet [24] demonstrated the ability to generate long-form, multi-instrument compositions across multiple genres using Transformer-based autoregressive decoding. Jukebox [25] further extended this paradigm to raw-audio generation, combining a hierarchical VQ-VAE representation with Transformer models to synthesize high-fidelity music directly in the waveform domain.

While Transformer-based models significantly improved musical coherence and diversity, they introduced substantial computational costs and inference latency, especially for audio-domain generation. To mitigate these challenges, more recent approaches have focused on diffusion-based generative models, which have achieved state-of-the-art performance in both symbolic and audio synthesis. Diffusion models operate by iteratively denoising random noise toward structured outputs, enabling stable training and high-quality generation. DiffRhythm [12], for example, employs latent diffusion to generate full-length songs with vocals in a single end-to-end framework, achieving impressive speed and fidelity. Similarly, Mustango [26] and AudioLDM2 [27] integrate diffusion processes with multimodal conditioning, allowing fine-grained control over timbre, rhythm, and semantic attributes through textual or symbolic prompts.

An important trend in modern generative music research is the increasing emphasis on controllability and semantic conditioning. Rather than generating music unconditionally, many systems incorporate external guidance signals such as text descriptions, emotional labels, or high-level musical attributes. Text-to-music models like MusicGen [10] exemplify this shift by enabling natural-language-conditioned music synthesis using a Transformer-based architecture trained on large-scale paired text-audio datasets. Such models allow users to specify genre, mood, instrumentation, or stylistic intent in intuitive linguistic terms, making generative music systems more accessible and interactive.

MidiNet [28] represents one of the earliest deep learning frameworks specifically designed for symbolic music generation using Generative Adversarial Networks (GANs). Unlike recurrent models that generate music sequentially, MidiNet formulates melody generation as a convolutional process over time, where each bar of music is generated conditioned on the previous bar and optional musical context such as chord progression. The generator

employs convolutional neural networks (CNNs) to produce piano-roll representations, while the discriminator evaluates their realism. By incorporating chord conditioning, MidiNet enables a degree of harmonic control, allowing the generated melodies to remain musically consistent with predefined harmonic structures. This design demonstrates that adversarial learning can be effectively applied to symbolic music generation and highlights the potential of conditional generation for controllable music synthesis. However, MidiNet [28] primarily focuses on monophonic melody lines and short-term structure, limiting its ability to model long-range dependencies and complex multi-instrument arrangements.

MuseGAN [23] extends the GAN-based paradigm to multi-track symbolic music generation, addressing one of the key limitations of earlier models like MidiNet. Instead of generating a single melody line, MuseGAN simultaneously produces multiple instrumental tracks-such as bass, drums, melody, and accompaniment-using a shared latent representation. The model introduces several architectural innovations, including temporal structure modeling at both bar-level and track-level, which helps maintain rhythmic consistency and inter-track coordination. MuseGAN demonstrates that GANs can learn hierarchical musical structure and inter-instrument relationships without explicit symbolic rules. Despite these advances, MuseGAN still operates within fixed-length segments and lacks explicit mechanisms for long-term musical form or semantic control. Additionally, as with many GAN-based models, training stability and mode collapse remain challenges, motivating later research to adopt autoregressive Transformers or diffusion-based approaches for improved scalability and controllability.

Collectively, these developments illustrate a clear progression: from symbolic rule-based systems to deep neural sequence models, and finally to multimodal, language-conditioned generative frameworks. This evolution highlights two key insights relevant to the present thesis. First, high-quality music generation increasingly depends on intermediate representations that capture semantic and structural intent rather than raw signal-level details alone. Second, natural language has emerged as a powerful and flexible modality for expressing musical concepts, enabling controllable and interpretable generation. These insights motivate the language-mediated design adopted in this study, where generative AI for music is integrated into a broader multimodal pipeline that bridges video understanding and audio synthesis through natural language.

2.2 Video-to-Music Generation

Video-to-music generation aims to automatically synthesize music that is temporally, semantically, and emotionally aligned with visual content. Unlike traditional music generation tasks that operate independently of visual input, video-to-music generation requires understanding dynamic visual cues, such as motion, scene changes, and narrative progression, and translating them into meaningful musical structures. This task is inherently multimodal and one-to-many: a single video can be accompanied by multiple musically valid soundtracks, each emphasizing different aspects of mood, rhythm or style. As a result, research in this area has evolved through several methodological stages, gradually shifting from low-level motion-driven mappings toward high-level semantic and language-mediated frameworks.

Early video-to-music systems primarily focused on mapping low-level visual features to musical parameters, under the assumption that visual motion intensity correlates with rhythmic energy. Foley Music, proposed by Gan et al. [1], is a representative example, where human pose keypoints extracted from video are mapped to MIDI sequences using Graph Neural Networks and Transformers. This approach enables interpretable connections between motion trajectories and melodic contours, making it suitable for domains such as dance or human activity videos. Similarly, D2M-GAN [2] employs vector-quantized audio representations and adversarial training to generate music conditioned on dance motion features. By directly coupling movement dynamics with audio synthesis, these systems achieve strong rhythmic correspondence.

However, motion-driven approaches are typically domain-specific and struggle to generalize beyond scenarios where motion is the dominant semantic signal. Videos with static scenes, abstract narratives, or emotional storytelling often lack explicit motion patterns that can meaningfully guide music generation. As a result, music generated by these methods may be rhythmically synchronized but semantically shallow, failing to capture narrative context or emotional nuance.

To overcome the limitations of purely motion-based conditioning, subsequent research introduced semantic alignment mechanisms that incorporate higher-level visual understanding. VMAS [4] represents a key milestone in this transition. VMAS employs contrastive learning to align video and music embeddings in a shared latent space, ensuring that semantically related video-music pairs are closer than unrelated ones. An autoregressive Transformer is then used to generate music conditioned on video semantics, allowing the model to capture both rhythmic and narrative correspondence.

This paradigm shift toward semantic conditioning is further reflected in survey works by Ji et al. [5] and Wang et al. [3], which categorize video-to-music systems based on conditioning strategies (motion-based vs. semantic-based), representation domains (symbolic vs. audio), and model architectures (direct mapping vs. cross-attention). These surveys highlight a growing consensus that high-level semantic understanding is essential for generating musically meaningful soundtracks for general-purpose videos.

Emotion-aware models also play an important role in semantic video-to-music generation. Video2Music [6] introduces an affective multimodal Transformer that explicitly models emotional states derived from visual cues, such as color tone, scene composition, and facial expressions. By conditioning music generation on emotion embeddings rather than raw motion features, the system produces background music that better matches the perceived mood of the video. Similarly, V2Meow [7] incorporates visual beat detection to enhance rhythmic synchronization, but remains limited in its ability to control musical style or emotional progression over long durations.

Recent advances in diffusion models have further improved audio quality and temporal smoothness in video-to-music generation. Diff-BGM [15] introduces a diffusion-based framework with segment-aware cross-attention, allowing the model to generate background music that adapts to different video segments while maintaining global coherence. By conditioning the diffusion process on temporally segmented video features, Diff-BGM achieves smoother transitions and higher perceptual quality than earlier autoregressive models. Despite their improved fidelity, diffusion-based video-to-music models often require substantial computational resources and large paired video-music datasets. This makes them difficult to deploy in interactive or web-based environments, where inference speed and memory usage are critical constraints. Furthermore, most diffusion-based systems operate as end-to-end black boxes, offering limited transparency or user control over the generative process.

A recent and emerging direction in video-to-music research involves leveraging large language models (LLMs) and vision-language models (VLMs) as semantic intermediaries. FilmComposer [8] exemplifies this trend by employing LLM-driven multi-agent systems to simulate professional film scoring workflows for silent video clips. Instead of directly mapping video features to audio signals, FilmComposer [8] first generates textual scene descriptions, emotional summaries, and compositional plans, which are then translated into music. This approach demonstrates that natural language can serve as an effective abstraction layer for reasoning about narrative structure and musical intent.

Similarly, Mustango [26] and related hybrid frameworks integrate visual cues, textual descriptions, and symbolic representations within diffusion pipelines, enabling richer contextual understanding. These language-mediated approaches offer several advantages: improved interpretability, enhanced controllability, and compatibility with text-to-music models such as MusicGen [10]. However, many existing systems rely on proprietary APIs or large-scale computational infrastructure, limiting reproducibility and accessibility.

In summary, video-to-music generation has evolved from motion-driven mappings to semantically grounded, multimodal systems that increasingly incorporate language as a central component. While early methods achieved strong rhythmic synchronization, they lacked narrative and emotional depth. Semantic and diffusion-based models improved alignment quality but introduced high computational costs and limited interactivity. Language-mediated approaches address these challenges by decoupling semantic reasoning from audio synthesis, enabling more flexible and interpretable pipelines.

Nevertheless, a clear research gap remains. Few existing systems provide an open-source, end-to-end framework that integrates scene-level video understanding, natural language mediation, and controllable music generation in a computationally efficient manner. Addressing this gap motivates the approach proposed in this thesis, which leverages natural language as an intermediate representation to bridge video semantics and music synthesis, balancing alignment quality, user control, and practical deployability.

2.3 Emotion-Conditioned Music Generation

Emotion-conditioned music generation aims to produce musical content that reflects or evokes specific affective states such as happiness, sadness, tension, or calmness. Instead of relying purely on stylistic imitation, these models condition the generation process on emotion-related cues derived from text, audio, or visual modalities. One representative work is EMOPIA [29], which introduced a multimodal pop-piano dataset annotated along the valence-arousal space. EMOPIA [29] enables models to learn how combinations of pitch, rhythm, and dynamics correspond to different emotional quadrants. Through emotion-aware conditioning, subsequent generative models can synthesize music that more accurately conveys intended affective expressions, rather than simply reconstructing style patterns.

Beyond dataset construction, other research focuses on directly encoding affective features into the music generation pipeline. For example, Generating Music with Emotions [30] proposed a transformer-based symbolic model conditioned on continuous emotion embeddings learned from listener ratings. By mapping emotions to latent vectors instead of discrete labels, their system

could generate smooth transitions between feelings such as “joyful → melancholic” while maintaining tonal coherence. This approach demonstrates that continuous emotional representations improve both perceptual realism and controllability in generative music systems.

To further enhance expressive control, MuseMorphose [31] introduced a variational sequence model that disentangles musical attributes such as rhythm, harmony, and dynamics into independent latent factors. Although MuseMorphose [31] was not originally designed solely for emotion generation, its latent interpolation mechanism allows manipulation of emotional intensity and style transfer between different musical pieces. The model’s latent space provides a flexible interface for emotion-guided composition, supporting nuanced variations without explicit emotion labels. Together, these works show that affect-conditioned music generation has evolved from discrete emotion classification toward continuous, interpretable representations, paving the way for multimodal applications where visual or textual cues can serve as emotional prompts.

2.4 Text-to-Music Generation

While emotion-conditioned models focus on affective control, another major branch of generative music research explores text-to-music generation, where natural language descriptions serve as the conditioning input. This approach enables intuitive interaction with generative systems, allowing users to describe desired musical moods, instruments, or genres in plain text. One of the pioneering works in this direction is MusicLM [11] a hierarchical generative model that maps textual descriptions to audio via multiple layers of discrete audio tokens. By leveraging a pre-trained audio-language model (MuLan [32]) for alignment and a cascade of coarse-to-fine token generators, MusicLM [11] produces long-term coherent audio clips that reflect both the content and style expressed in the prompt. Its success demonstrated that language models can serve as powerful intermediaries for music generation when trained on large-scale text-audio pairs.

Similarly, MusicGen [10] proposed a transformer-based auto-regressive decoder that synthesizes high-fidelity music directly in the waveform domain. Unlike diffusion-based or multi-stage tokenizers, MusicGen [10] employs a single forward pass over quantized EnCodec tokens [33], resulting in faster inference with consistent rhythmic and harmonic structure. Moreover, by conditioning on textual captions or melodic examples, MusicGen [10] supports hybrid generation that combines both semantic and structural control, making it suitable for real-time or web-based applications.

In parallel, several open-source models aim to bridge symbolic and audio-level representations for controllable composition. MusTango [26] introduces

a multi-band diffusion architecture for text-to-music generation, emphasizing open accessibility and integration with symbolic representations such as MIDI. Complementing these efforts, MT3 [34] presents a multi-task transformer for music transcription that converts audio to symbolic notation across diverse instruments and datasets. This symbolic bridge allows bidirectional flow between text, MIDI, and waveform domains, facilitating more interpretable and modular music generation pipelines. Collectively, these works demonstrate the growing convergence between natural language processing and music generation, motivating architectures that use language as an intermediary modality - an approach directly aligned with the goals of this thesis.

2.5 Singing Voice and Lyric Generation

An essential subdomain of music generation involves producing singing voices and lyrics that complement instrumental or symbolic compositions. This task combines text generation, melody synthesis, and voice modeling - each requiring temporal alignment between linguistic and musical elements. Early systems relied on rule-based mapping between lyrics and notes, but recent advances in generative modeling have enabled more expressive and controllable synthesis of singing voices conditioned on melody, lyrics, or emotional cues.

A representative modern framework is DiffSinger [35], which introduced a diffusion-based architecture for singing voice synthesis. DiffSinger [35] models the denoising process of mel-spectrograms, allowing it to capture fine-grained temporal and timbral details of human singing. By conditioning on both phoneme-aligned lyrics and pitch contours, the model can generate realistic and expressive vocals consistent with given melodies. This diffusion-based paradigm has since become the foundation for high-fidelity singing synthesis due to its robustness against over-smoothing and its ability to control expressiveness across different singers or styles.

Another relevant work is SingSong [36], developed by Google Research, which performs zero-shot singing voice generation from instrumental audio and textual lyrics. SingSong [36] leverages an audio language model trained with AudioLM [37] to model long-term temporal dependencies between singing and accompaniment. The model generates high-quality singing voices directly in the waveform domain, conditioned on textual lyrics and background music, without requiring singer-specific data. This represents an important step toward open-domain singing synthesis compatible with language-based conditioning.

2.6 Music understanding of LLMs

Recent advances in Natural Language Processing (NLP) have greatly enhanced computational musicology and music information retrieval by enabling symbolic, structural, and semantic analysis of music through language models. Rather than directly generating sound, NLP-based systems interpret and reason about music's theoretical properties, such as tonality, harmony, rhythm, and form-through text representations. Early research integrated symbolic analysis with textual annotations, allowing neural models to predict chord progressions or genre labels from score data. More recent work, such as MusicBERT [21] applies Transformer architectures to encode symbolic sequences for downstream tasks including chord recognition, genre classification, and structural segmentation. These models show that linguistic-style embeddings can capture high-level music-theoretical features, providing a bridge between symbolic music and natural language descriptions.

Large language models (LLMs) have further expanded the scope of music analysis by learning to describe, critique, and reason about musical elements. Systems such as ChatMusician [20] and MuseCoco [38] extend pre-trained LLMs with domain-specific musical corpora, enabling text-based interpretation of melody, rhythm, and harmony. These models can explain chord functions, identify stylistic traits, and recommend compositional adjustments using natural language. This convergence of symbolic representation learning and textual reasoning highlights NLP's ability to act as a cognitive intermediary that interprets musical content beyond raw acoustics.

Complementing this, ChatMusician [20] demonstrates how large language models (LLMs) can handle symbolic music reasoning, lyric composition, and natural-language-conditioned music understanding. By unifying symbolic token representations (notes, chords, and lyrics) with text embeddings, ChatMusician [20] bridges natural language and structured music representations. This capability makes it ideal for systems where user prompts are expressed in free-form language, allowing automatic generation of both lyrics and corresponding musical features such as tempo, harmony, and emotion.

Together, these studies illustrate a clear trajectory: from specialized singing synthesis systems toward language-integrated, multimodal frameworks that can generate coherent lyrics, melody, and expressive vocal timbre in a unified process. Such systems provide the foundation for the thesis goal - building an open-source web application that uses natural language as the intermediary between video understanding and music creation.

2.7 Discussion

The review of existing literature shows that while text-to-music systems such as MusicGen [10] achieve impressive quality and controllability, they are limited to textual prompts and lack the ability to account for dynamic, multimodal cues present in videos. Video-to-music systems like D2M-GAN [2] and VMAS [4] have advanced synchronization between visual motion and audio rhythm, yet they often struggle with semantic richness: the generated music may follow the beat but fail to capture the intended mood, narrative, or emotional tone of the scene. Other projects such as Diff-BGM [15] have pushed output quality higher through diffusion techniques but remain computationally heavy, limiting their deployment in real-time or web-based environments.

Another limitation across many systems is interactivity. Most current projects operate in a one-shot manner, where the user provides input (either video or text) and receives an output without the opportunity for refinement. This reduces creative control, which is especially important in artistic domains. Furthermore, while some models are open-source, many cutting-edge systems remain tied to proprietary APIs, restricting accessibility for researchers and developers. Even when source code is available, the lack of unified integration - combining video understanding, language mediation, music generation, and lyric alignment - means that users must work with fragmented tools rather than a cohesive framework.

These limitations highlight a clear research gap. A successful system should not only generate high-quality music but also integrate semantic video understanding, natural language interaction, and lyric generation in a single pipeline. It should remain lightweight enough for deployment in a web-based environment while retaining the openness and flexibility of community-driven AI models. Addressing these limitations is the central motivation of the proposed thesis project.

CHAPTER 3: PROPOSED APPROACHES

This chapter describes the proposed end-to-end pipeline for generating music from video using natural language as an intermediate representation. It details the overall system architecture and explains each processing stage, including scene segmentation, video captioning, music feature inference, prompt preparation, music generation, and output rendering. The chapter emphasizes the design choices that improve semantic alignment, interpretability, and suitability for web-based deployment using open-source models.

3.1 Pipeline overview

This study proposes an end-to-end open-source pipeline for generating music from videos, integrating multiple artificial intelligence models across vision, language, and audio domains. The methodology is designed to ensure semantic alignment between video content and generated music while allowing user interaction through natural language. The pipeline consists of seven main stages: (1) scene detection, (2) scene segmentation, (3) video captioning, (4) music feature proposal, (5) prompt preparation, (6) music generation, and (7) output rendering.

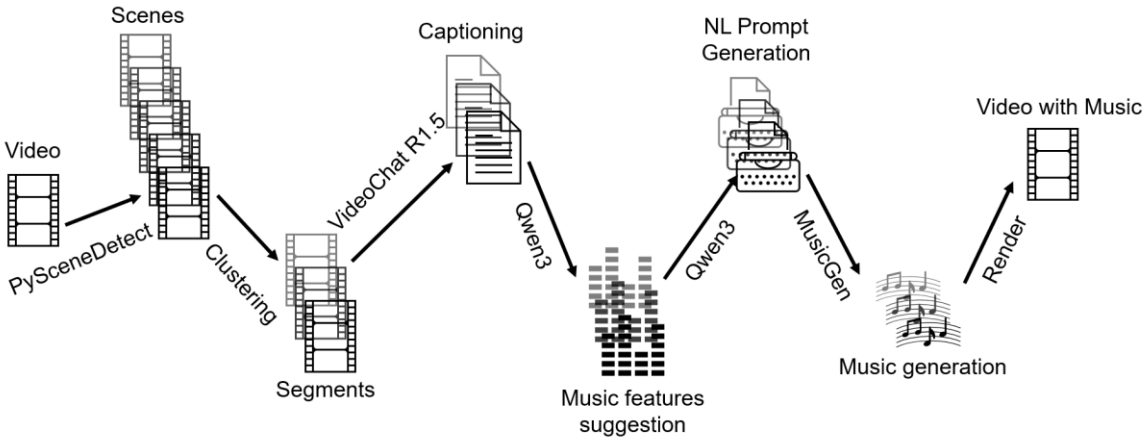

Figure 1: Pipeline overview

Figure 1 presents the overall architecture of the proposed video-to-music generation pipeline. The system is designed as an end-to-end open-source framework that integrates vision, language, and audio models, with natural language acting as the intermediate representation to ensure semantic alignment between video content and generated music. Given an input video, scenes are first detected and clustered into semantically coherent segments using PySceneDetect [39]. Each segment is then described using a pretrained video captioning model to extract high-level semantic information. Based on these captions, a large language model infers suitable musical attributes, which are converted into natural language prompts for music generation. Background

music is subsequently generated using MusicGen [10] with a continuation strategy to maintain temporal coherence, and finally rendered together with the video to produce the output.

3.2 Video Upload and Scene Segmentation

The first stage of the proposed pipeline aims to divide an input video into semantically meaningful scenes, that can later be processed independently. Scene detection is critical for maintaining contextual relevance in music generation, as changes in visual content (e.g., mood, lighting, motion) often correspond to transitions in musical intensity or emotion.

To achieve this, the system employs a hybrid approach combining PySceneDetect [39] and CLIP-based semantic clustering. PySceneDetect [39] is used to identify preliminary scene boundaries by analyzing frame differences and histogram variations, effectively splitting the video into smaller segments based on visual discontinuities. However, purely visual change detection may not always align with semantic or narrative transitions. To address this limitation, a scene embedding and clustering mechanism is implemented using OpenAI's CLIP model [40], which encodes each scene's visual content into a high-dimensional semantic vector representation.

For each detected scene, a fixed number of frames (typically six) are sampled uniformly across the segment using OpenCV [41] library. These frames are processed through CLIP's image encoder to obtain feature embeddings that represent the overall visual semantics of that scene. The embeddings are normalized and averaged to form a single feature vector per scene. Once embeddings for all scenes are computed, a cosine similarity-based adjacency clustering algorithm merges consecutive scenes that are semantically similar and temporally adjacent, subject to a maximum duration threshold (e.g., 50 seconds). This ensures that scenes with consistent visual themes such as continuous dialogue or similar environmental context will be grouped together, while maintaining manageable segment lengths for subsequent processing.

Algorithm 1: Conditional Adjacency Clustering
1: **Given** n_{scene}: number of scenes E_i: embedding vector of scene i-th Dur_i: duration of scene i-th Dur_{max}: maximum duration of each cluster 2: **Initialize** each scene for each clusters, distance between scene i-th and (i+1)-th $Distance_i = 1 - CosineSimilarity(E_i, E_{i+1})$ 3: **While** min(Distance) != ∞ and $n_{cluster} > 1$: 4: i = position of the minimum in Distance array (2 closest clusters) 5: **If** $Dur_i + Dur_{i+1} <= Dur_{max}$ then: 6: Merge cluster (i+1)-th into cluster i-th by recalculate cluster embedding, scenes, end time and duration. 7: $Ei = \frac{Dur_i * E_i + Dur_{i+1} * E_{i+1}}{Dur_i + Dur_{i+1}}$ (recalculate cluster embedding) 8: Delete cluster (i+1)-th and $Distance_i$ because already merged 9: Recalculate $Distance_i$, $Distance_{i+1}$ and $Distance_{i-1}$ 10: **Else**: 11: Setting Distance of cluster idx and (idx + 1)-th to ∞ for seperation of these scenes

Algorithm 1 presents the proposed Conditional Adjacency Clustering method, designed to merge semantically similar and temporally adjacent video scenes into coherent segments. The algorithm begins by initializing each detected scene as an individual cluster, where each scene E_i is represented by its CLIP embedding vector, and its duration Dur_i is recorded. The pairwise distances between adjacent scenes are computed using cosine dissimilarity, defined as $Distance_i$. During iteration, the algorithm identifies the pair of clusters with the smallest distance value - indicating the highest semantic similarity, and checks whether their combined duration remains below the predefined maximum threshold Dur_{max} (e.g., 50 seconds). If the condition is satisfied, the two clusters are merged by recalculating a new cluster embedding as the duration-weighted average of their feature vectors. The merged cluster inherits all scene information, including start and end timestamps, updated duration, and unified embeddings. Distances between adjacent clusters are then recalculated to maintain consistency. When a potential merge would exceed the duration limit, the corresponding distance is set to infinity to prevent further merging. This iterative process continues until no eligible clusters remain,

resulting in a set of semantically homogeneous and temporally constrained segments. By combining similarity-based merging with duration control, this algorithm preserves both narrative flow and contextual coherence - ensuring that each resulting segment serves as a meaningful unit for subsequent video summarization and music generation stages.

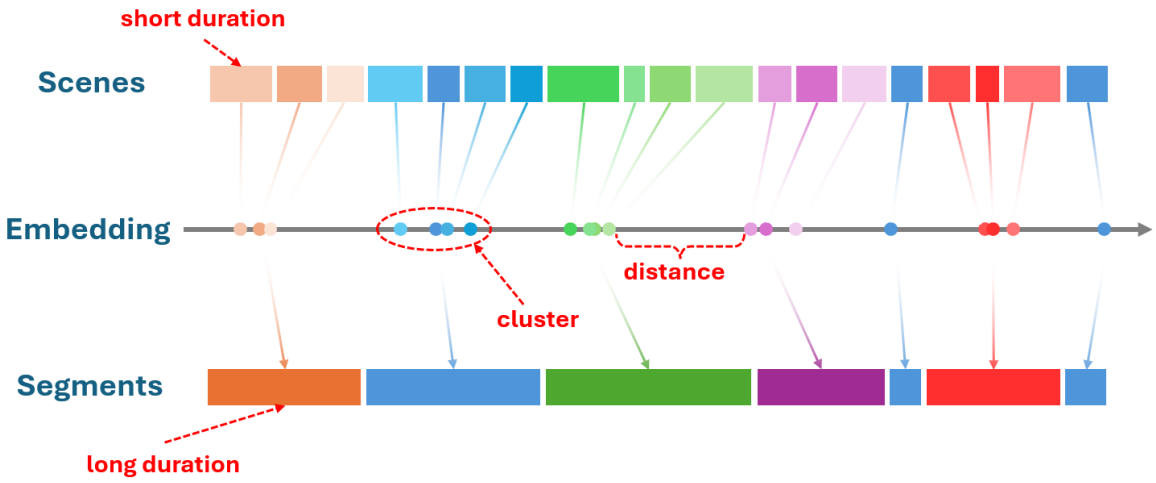

Figure 2: Example of clustering algorithm

3.3 Video Captioning

After merging semantically coherent scenes into clusters, each resulting video segment is annotated with natural language descriptions using a multimodal large language model (MLLM). In this study, the VideoChat-R1.5 [42] model from OpenGVLab - an enhanced variant of the Qwen2.5-VL [43] architecture - was employed for dense video captioning. This model integrates visual and linguistic reasoning, allowing it to interpret spatiotemporal dynamics, object interactions, and emotional tone within video sequences. To improve inference efficiency and reduce GPU memory usage, the model was loaded with 4-bit quantization using the BitsAndBytes configuration, maintaining computation in bfloat16 precision for numerical stability.

Each clustered video segment was passed to the model alongside a structured instruction prompt that directed the system to produce expressive, scene-level summaries. The prompt explicitly asked the model to describe the visual content, actions, emotional atmosphere, and motion cues, producing captions of approximately 5 - 8 sentences per segment. The model then generated dense captions through an autoregressive decoding process, conditioned jointly on visual and textual inputs.

The resulting captions were serialized into a JSON structure containing metadata for each cluster, specifically, start and end timestamps, duration, file path, and generated description. This structured output enables downstream modules to retrieve semantically relevant textual summaries that guide music

feature proposal and lyric generation stages. By leveraging a high-capacity vision-language foundation model, this step bridges the visual and linguistic domains, effectively transforming low-level visual motion cues into semantically interpretable narratives that inform the subsequent multimodal generation process.

3.4 Music Feature Proposal

After obtaining semantic video captions, the system employs a Small Language Model (SLM) to infer high-level musical attributes that guide the subsequent music generation process. This stage focuses on designing and fine-tuning an improved variant of ChatMusician, which is responsible for translating textual video summaries into structured music-theoretical features.

3.4.1 Limitations of the Original ChatMusician

The original ChatMusician model [20], based on LLaMA2 [44], demonstrated the feasibility of aligning large language models with symbolic music understanding through supervised fine-tuning on the MusicPile dataset [34]. However, its effectiveness remains constrained by several limitations documented in its original study:

- Restricted model capacity: LLaMA2-7B [44] exhibits limited contextual retention and struggles with long-sequence reasoning, especially for extended compositional structures such as chord progressions or orchestral arrangements.
- Limited multilingual and cross-domain alignment: The model shows weaker performance when handling mixed-language prompts, non-Western music traditions, or multimodal contexts (e.g., emotion-based or visual prompts).
- Inconsistent feature granularity: While capable of generating basic attributes like tempo or key, the model often fails to produce coherent high-level features such as dynamics, texture, and stylistic intent.
- Weak instruction-following ability and hallucination tendency - ChatMusician [34] frequently misinterprets user instructions or generates inconsistent feature sets unrelated to the prompt context. It sometimes “hallucinates” special when suggest about chord progression

To overcome these shortcomings, this research will use Qwen3-4B-Instruct-2507 [45], a state of the art open-source multilingual SLM developed by Alibaba Cloud replace with ChatMusician [20] for music feature suggestion step.

3.4.2 Music features suggestion

After the captioning process, each scene cluster is associated with a natural language description generated by VideoChat-R1.5 [42], as discussed in Section 3.3 Video Captioning. Based on this description, the Qwen3-4B-Instruct-2507 model [45] is employed to infer a structured set of high-level musical attributes that are relevant to music composition and soundtrack design. Instead of directly generating music, the model acts as a semantic intermediary that translates visual and narrative cues into interpretable musical concepts. The extracted music features include:

Genre: The stylistic or cultural category that defines the overall musical tradition or production style. Genre encapsulates historical periods (e.g., Baroque, Romantic), popular idioms (e.g., Pop, Rock, Jazz, Hip Hop), electronic sub-genres (e.g., House, Ambient, Trance), and media-oriented forms such as Film Score or Synthwave. This attribute provides a coarse but essential constraint on instrumentation, rhythm, and production aesthetics.

Mood: The emotional or affective quality of the music, reflecting the psychological response evoked by the visual scene. Mood plays a critical role in shaping expressive musical parameters such as perceived energy, harmonic tension, and articulation. This feature ensures emotional congruence between the video content and the generated soundtrack.

Style: Style captures the compositional and production characteristics that refine how the genre and mood are realized in practice. Unlike genre, which defines what musical tradition is used, style specifies how the music is performed or produced. Examples include cinematic, orchestral, minimal electronic, lo-fi, ambient-driven, rhythmic action cue, or modern hybrid score. This attribute controls aspects such as arrangement density, rhythmic emphasis, layering, and overall sonic texture, making it particularly important for aligning the generated music with film-scoring conventions.

Instruments: This feature specifies the primary sound sources or timbral agents used in the composition, typically limited to one or two dominant instruments (e.g., piano, guitar, violin, synthesizer, drums). Instrument selection strongly influences the perceived tone color and genre identity, and serves as a direct cue for the downstream music generation model.

Context: Context provides a short natural-language summary describing the intended cinematic, emotional, or narrative purpose of the music. Rather than encoding low-level musical details, this field captures high-level semantic intent, such as whether the music supports tension, action, introspection, or resolution within the scene. Context acts as a semantic bridge between visual storytelling and musical interpretation.

All attributes are intentionally constrained to short, high-impact descriptors and exclude low-level musical parameters such as tempo (BPM), key, harmony, or chord progressions. This design choice avoids over-specification and allows the generative model to retain creative flexibility while remaining semantically grounded. Overall, this structured features providesan interpretable and modular musical representation that can either be consumed directly by the music generation model or manually edited by users. By separating semantic understanding from audio synthesis, the system improves controllability, transparency, and extensibility in video-to-music generation.

3.5 Prompt preparation

After the structured music features are inferred, a second prompt-generation stage is applied to convert these attributes into a concise, MusicGen-compatible natural language prompt. While the extracted features are interpretable and modular, MusicGen [10] performs best when guided by short descriptive prompts that emphasize auditory characteristics rather than abstract labels. To address this, the Qwen3-4B-Instruct-2507 [45] model is reused as a prompt refiner. Given the inferred Genre, Mood, Style, Instruments, and Context, the model rewrites the information into a compact, one-to-two-sentence description following a set of strict constraints:

1. The prompt must explicitly describe rhythmic energy (e.g., pulsing rhythm, driving beat, layered textures).
2. Exactly one clear energy cue must be included to guide musical intensity.
3. Metaphorical or poetic language is prohibited to avoid ambiguity.
4. Terms implying silence, minimalism, or inaudibility (e.g., soft, faint, subtle, distant) are disallowed unless explicitly required by the scene.
5. The final output must resemble real MusicGen [10] prompt examples used in prior demonstrations and documentation.

This controlled rewriting process transforms abstract musical features into a form that is both expressive and compatible with the generative constraints of MusicGen [10]. By enforcing stylistic consistency and eliminating ambiguous language, the prompt preparation stage significantly improves the stability and relevance of the generated audio.

Importantly, this two-stage prompting strategy, feature inference followed by prompt refinement, decouples semantic reasoning from audio generation. As a result, the system remains model-agnostic and can be adapted to other text-to-music frameworks with minimal modification, reinforcing the flexibility of the proposed web-based video-to-music generation pipeline.

Music features suggestion	Music generation prompt preparation
Genre: “Sci-fi thriller” Mood: “Foreboding, tense” Style: “Minimalist, atmospheric” Instruments: “Synths, ambient drones, subtle percussion” Context: “From mystery to revelation”	“A tense, pulsing electronic score with deep synths and low-frequency drones, featuring a steady, driving rhythm and subtle percussion.”
Genre: “Ambient Electronic” Mood: “Reflective, Introspective” Style: “Minimalist, Atmospheric” Instruments: “Pulsing synth, ambient pads” Context: “Urban solitude, quiet determination”	“A minimalist ambient electronic track with a steady pulsing synth and atmospheric pads, featuring a calm yet focused energy.”
Genre: “Action” Mood: “High-energy” Style: “Electronic pulse” Instruments: “Synths, drums, bass” Context: “Racing and athletic intensity”	“A high-energy electronic pulse with driving drums, pulsing synths, and deep bass, capturing the intensity of racing and athletic motion.”

Table 1: Some examples of music features and music generation prompts

3.6 Music Generation

After the MusicGen-compatible prompts are prepared as described in Section 3.5 Prompt preparation, the system proceeds to the audio synthesis stage. This stage is responsible for generating continuous background music that aligns temporally and semantically with the video content. The implementation relies on MusicGen [10], an open-source text-to-music model released as part of Meta’s AudioCraft framework.

3.6.1 Segment-Based Music Generation

Given that the input video is decomposed into multiple scene clusters with heterogeneous durations, a naive single-pass music generation approach is insufficient to ensure both temporal coverage and musical coherence. In particular, the MusicGen [10] model imposes an upper bound on generation

length per forward pass, and abrupt regeneration across segments may introduce audible discontinuities. To address these limitations, we propose a segment-based music generation strategy with explicit continuation handling and temporal alignment**.**

For each video cluster, the target music duration is computed to exactly match the temporal length of the corresponding video segment. Music generation is performed sequentially across clusters, allowing the output of the previous cluster to influence the next one. Specifically, a fixed overlap window of 10 seconds is used to maintain musical continuity between adjacent clusters. Two generation regimes are considered depending on the cluster duration:

Short segments (≤ 30 seconds): When the cluster duration does not exceed the maximum reliable generation length of MusicGen [10], music is generated in a single pass. If the cluster is not the first one, the system loads the final 10 seconds of the previous cluster's audio and applies MusicGen's generate_continuation function. The conditioning segment is then removed from the generated output to avoid duplication, and the resulting audio is trimmed to the exact target duration. This ensures smooth transitions while preserving precise temporal alignment with the video.

Long segments (> 30 seconds): For longer clusters, a two-pass generation strategy is adopted. In the first pass, music is generated either from scratch or as a continuation of the previous cluster, producing an initial segment of up to 30 seconds. The last 10 seconds of this output are then extracted and used as conditioning input for a second continuation pass. The overlapping conditioning region is removed from the second output, and the two generated segments are concatenated to form a single coherent audio track. Finally, the combined audio is cropped to match the desired cluster duration exactly. This approach allows the system to handle arbitrarily long video segments while respecting the model's generation constraints.

Overall, this segment-based generation and alignment strategy enables scalable music generation for long-form videos, preserves temporal coherence across scene boundaries, and provides a stable foundation for downstream evaluation of video-music alignment quality.

3.6.2 Music Continuation Across Clusters

To maintain musical coherence across scene boundaries, the system employs MusicGen's continuation generation capability. Instead of generating each cluster's music independently, the last 10 seconds of the previous audio cluster are used as conditioning input for the next generation. Concretely, the waveform corresponding to the final five seconds of the preceding cluster is extracted and provided to the model as a continuation prompt. The newly generated audio discards the overlapping conditioning window, ensuring that

only novel content is appended. This process produces smooth transitions between consecutive clusters and avoids abrupt changes in timbre, rhythm, or energy.

For long segments, continuation is applied both across clusters and within a cluster itself when multi-pass generation is required. The final waveform is obtained by concatenating the generated segments and trimming the result to match the exact duration of the video cluster. This continuation-based design is essential for preserving musical flow in multi-scene videos and approximates the behavior of human-composed film scores, where musical ideas evolve gradually rather than resetting at scene cuts.

3.6.3 Loudness Normalization

In the music feature suggestion stage, all generated music samples are normalized to a fixed loudness level (−14 dB) before feature extraction. This step ensures that differences between music samples are caused by their musical content rather than by variations in overall volume. Loudness normalization is necessary because many audio features, such as spectral descriptors, zero-crossing rate, harmonic distortion, and beat detection, are highly sensitive to signal amplitude. Without normalization, louder audio segments may produce artificially higher feature values, leading to biased or unfair evaluation results when comparing different models or generation strategies.

The value of -14 dB is chosen as a practical reference level commonly used in video and online media content. It is sufficiently loud to preserve rhythmic and spectral details while avoiding excessive amplification that could introduce distortion. This level also helps maintain consistent perceived loudness across consecutive music segments, making it well suited for video-conditioned music generation and evaluation.

3.7 Output Rendering

After music has been generated for each scene cluster as described in Section 3.6 Music Generation, the final stage of the pipeline focuses on reconstructing a single coherent output video. This step integrates the individually generated audio segments into a continuous soundtrack and synchronizes it with the original full-resolution video. The goal of this stage is to ensure smooth auditory transitions, temporal alignment, and high-quality final rendering suitable for end-user consumption.

Each scene cluster produces an individual audio file (music_cluster_i.wav) corresponding to its video duration. These files are stored in a designated directory and represent the sequential musical structure of the video. To preserve the original narrative order, the audio files are

automatically sorted based on their cluster indices. This ensures that the temporal order of the generated music strictly follows the chronological progression of the video scenes. The system loads all audio segments as AudioFileClip objects, which allows flexible manipulation of timing and overlap during composition.

Direct concatenation of audio segments can introduce perceptually noticeable discontinuities at scene boundaries. To address this issue, a crossfade-based composition strategy is employed. For each consecutive pair of audio segments, a short overlap window of fixed duration (1.5 seconds in the current implementation) is applied. The next audio segment is scheduled to start slightly before the end of the previous one, allowing the two waveforms to overlap and blend smoothly. This process is implemented using CompositeAudioClip, which mixes overlapping segments in the time domain. The resulting soundtrack exhibits gradual transitions in rhythm, timbre, and energy, closely resembling professionally edited film soundtracks. The total duration of the final audio track is dynamically computed based on the combined lengths of all clusters. Rather than rendering music on pre-cut video segments, the system reloads the original full-resolution video file during the output rendering stage. This design choice avoids cumulative encoding artifacts and preserves the original visual quality.

Once the final audio duration is determined, the full video is temporally trimmed to match the length of the generated soundtrack. This guarantees precise synchronization between audio and video, preventing silent tails or desynchronized endings. The composed audio track is then attached to the video using MoviePy's audio-video binding mechanism. The final output video is rendered using standard, widely supported codecs to ensure compatibility across platforms and browsers. Specifically, H.264 (libx264) is used for video encoding, while AAC is employed for audio compression. Frame rate settings are inherited from the original video to maintain visual consistency. The result is a single video file that contains the original visual content augmented with automatically generated, semantically aligned background music.

This rendering step represents the culmination of the proposed pipeline, transforming raw video input into a fully produced multimedia artifact. The output rendering stage plays a critical role in bridging generative AI models with real-world multimedia applications. By combining continuation-aware music generation with crossfade-based audio composition, the system avoids abrupt transitions and maintains perceptual coherence throughout the video.

Furthermore, the decision to reattach audio to the original full-resolution video ensures that the generative process does not degrade visual fidelity. The modular structure of this stage allows future extensions such as user-controlled

fade durations, dynamic re-rendering of individual clusters, or interactive preview within a web interface.

3.8 Evaluation metrices

Evaluating a video-to-music generation system is challenging because it involves two intertwined objectives: (1) generating musically plausible and perceptually stable audio, and (2) ensuring meaningful semantic and temporal alignment between the generated music and the visual content. Subjective listening tests are often expensive and difficult to reproduce, especially in an academic setting. Therefore, this work adopts an objective, reproducible, and code-driven evaluation framework, grounded in signal processing and multimodal representation learning. Following prior work in video-to-music generation and cross-modal alignment (e.g., Diff-BGM [15], V2Meow [7]), the proposed evaluation protocol is divided into two complementary groups of metrics:

- **Music Quality Metrics**: assess intrinsic acoustic properties of the generated music.
- **Video-Music Relationship Metrics**: quantify semantic, rhythmic, and temporal alignment between video and music.

All metrics are computed automatically using open-source libraries, ensuring reproducibility and scalability. In video-to-music generation, directly comparing the generated music to the ground-truth music is often inappropriate and potentially misleading. Unlike tasks such as speech synthesis or audio reconstruction, the video-to-music problem is inherently one-to-many: a single video can be plausibly accompanied by multiple musically valid soundtracks that differ in melody, instrumentation, rhythm, or harmony while still being perceptually appropriate. As a result, low similarity between generated music and the ground-truth audio does not necessarily imply poor generation quality.

Instead, we evaluate the consistency of the video-music relationship. Specifically, we compare how music relates to the video in the generated pair versus how music relates to the video in the ground-truth pair. By focusing on cross-modal alignment, such as semantic correspondence, temporal synchronization, and structural consistency, we assess whether the generated music captures the same conditioning behavior imposed by the video, rather than attempting to replicate a single reference soundtrack. This relational evaluation better reflects the true objective of video-to-music generation and provides a more faithful measure of model performance.

3.8.1 Music quality

Music quality metrics evaluate the audio signal independently of the video, focusing on stability, timbral characteristics, and noise artifacts. Since the system generates background music rather than symbolic scores, waveform-based features are preferred over note-level accuracy.

(a) Total Harmonic Distortion (THD): used to quantify the spectral purity of the generated audio by measuring the amount of harmonic energy relative to the fundamental frequency. Given an audio waveform, we first compute its frequency spectrum using the real-valued Fast Fourier Transform (FFT) from NumPy library [46] and identify the fundamental frequency as the spectral component with the maximum magnitude. Based on the detected fundamental frequency, THD is computed by accumulating the energy of its higher-order harmonics up to a predefined number of harmonics. The final THD score is defined as the ratio between the root-sum-square energy of these harmonics and the magnitude of the fundamental component. A lower THD value indicates cleaner harmonic structure and reduced spectral distortion in the generated music.

(b) Spectral Flatness (SF): measures how noise-like or tone-like an audio signal is by comparing the geometric mean to the arithmetic mean of its spectral magnitude. Given the audio waveform, we compute its short-time Fourier transform (STFT) and estimate spectral flatness for each time frame using standard audio analysis tools. The final metric is obtained by averaging spectral flatness values over time. Lower values indicate more harmonic, tonal structure, while higher values suggest noisier or more broadband characteristics in the generated music.

(c) Spectral Centroid Variability (SCV): measures the temporal variability of the spectral centroid, which reflects the perceived brightness of an audio signal. Based on the input waveform, the spectral centroid is computed for each time frame using standard audio feature extraction methods using Librosa library [47]. The standard deviation of the centroid values over time is then used as the metric, where higher variability indicates greater spectral dynamics and timbral variation in the generated music.

(d) Zero-Crossing Rate (ZCR): quantifies the noisiness of an audio signal by measuring the rate at which the waveform crosses the zero-amplitude axis. It is computed directly from the audio waveform and averaged over time. Higher ZCR values typically indicate more high-frequency or noise-like components, whereas lower values correspond to smoother, more tonal audio signals. Both SCV and ZCR are calculated using Librosa library [47].

3.8.2 Video-Music relationship

Music quality alone does not guarantee suitability for video accompaniment. Therefore, the second group of metrics evaluates cross-modal alignment, capturing semantic correspondence, rhythmic synchronization, and structural coherence between video and music. To achieve modality-agnostic comparison, this work uses ImageBind [48], a unified multimodal embedding model that maps vision and audio into a shared latent space. To evaluate the semantic and temporal correspondence between video content and generated music, we adopt a set of embedding-based cross-modal metrics:

$$V = \{v_t | t = 1, 2, \dots, N\}$$
$$A = \{a_t | t = 1, 2, \dots, N\} \quad (1)$$

While N is the number of sliding window across video timeline with 3 seconds window size and 1 seconds overlap size, v_t is the video embedding of sliding window t-th calculated by average of frame embeddings using ImageBind [48] in that window duration, a_t is the audio embedding of sliding window t-th using ImageBind [48].

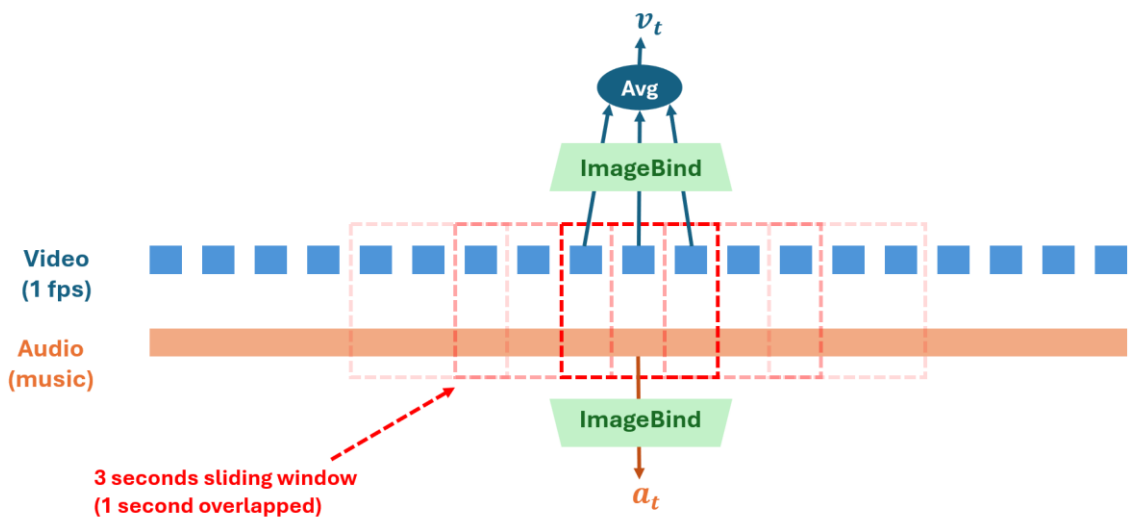

Figure 3: Video and Audio embedding method for evaluation

(a) Cross-Modal Embedding Correlation (CMEC): measures semantic alignment between video and music, embeddings are extracted using ImageBind [48], a unified multimodal representation model capable of embedding vision and audio into a shared latent space. Given sequences of audio and video embeddings, we first compute the mean embedding for each modality to obtain a single global representation. These mean embeddings are then $\ell 2$-normalized, and their cosine similarity is computed as the final score. A higher correlation value indicates stronger global semantic alignment between the generated music and the video content.

$$CME = \frac{\bar{v}^T \bar{a}}{\|\bar{v}\|_2 \|\bar{a}\|_2} \quad (2)$$

While $\bar{v}$ and $\bar{a}$ are average vector embedding of list V and list A respectively discussed above, $\|x\|_2$ is $\ell 2$-normalized function. This formular is the cosin similarity between normed mean embedding of audio and video.

(b) Beat-Motion Synchrony (BMS): evaluates temporal alignment between rhythmic events in the music and motion intensity changes in the video. Visual motion is estimated from frame-wise absolute differences between consecutive video frames and then aggregated over time. Instead of using individual frames, the motion signal is grouped into sliding temporal windows of 3 seconds with an overlap of 1 second. For each window, motion energy is computed as the average motion magnitude within that interval, providing a smooth representation of how strongly the video content moves over time. Musical beats are extracted from the audio signal as time-stamped events. Using the same overlapping temporal windows as for motion energy, beat density is calculated by counting the number of beats occurring within each window. This results in a time-varying measure of rhythmic activity, where higher values indicate denser or faster musical rhythms. To measure beat-motion synchrony, the motion energy and beat density sequences are independently normalized to remove scale differences. Synchrony is then quantified by computing the Pearson correlation between the two normalized sequences. A higher correlation indicates that increases in visual motion are consistently accompanied by increases in musical beat density, reflecting stronger audiovisual rhythmic alignment.

(c) Cross-Modal Similarity Alignment (CMSA): measures how uniformly audio-video similarity is distributed over time. The video and audio streams are first segmented into overlapping temporal windows of 3 seconds with an overlap of 1 second, producing two sequences of window-level embeddings. A cross-modal cosine similarity matrix is then computed, where each entry represents the similarity between a video segment and an audio segment at corresponding temporal windows. This matrix captures how well visual and auditory content match across time. When video and music are temporally and semantically aligned, high similarity values are expected to concentrate along the main diagonal of the similarity matrix, as diagonally aligned entries correspond to video and audio segments covering the same temporal intervals. We therefore evaluate alignment by averaging the similarity values along the main diagonal (or within a narrow diagonal band to allow small timing deviations). A higher diagonal alignment score indicates stronger video-music correspondence over time.

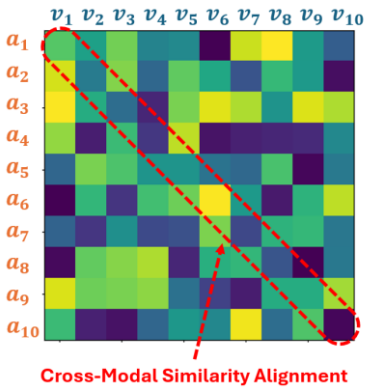

Figure 4: Example of similarity matrix between audio and video

(**d) Self-Similarity Matrix (SSM) Alignment**: evaluates whether the internal temporal structure of the generated music is consistent with that of the video. Given sequences of video and audio embeddings with equal temporal resolution, we first construct self-similarity matrices for each modality using cosine similarity, where each matrix captures pairwise similarity relationships between all temporal segments within the same modality. Based on these self-similarity matrices, we compute the Pearson correlation between their upper triangular elements (excluding the main diagonal). This comparison evaluates whether relative similarity patterns across time are preserved across modalities. A higher SSM-Alignment score indicates that the temporal structure of the generated music closely reflects the structural evolution of the video content, beyond direct frame-level synchronization.

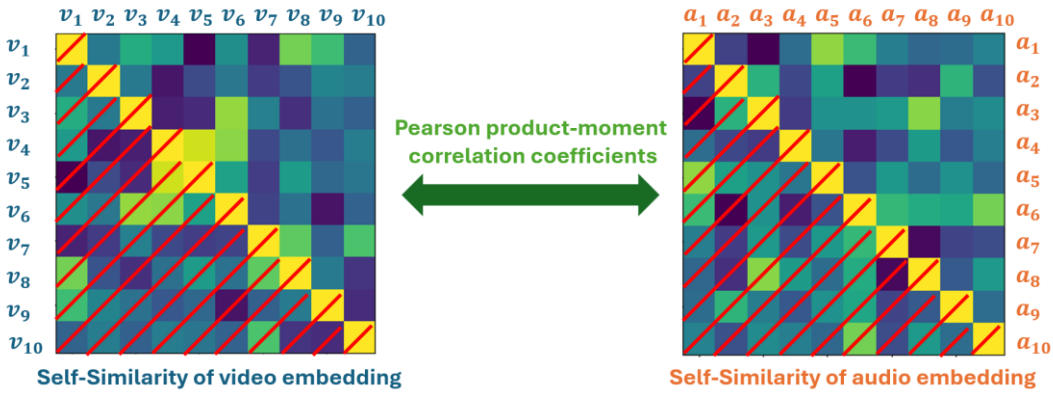

Figure 5: Example of Self-Similarity Matrix Alignment metric

Together, these metrics form a comprehensive, multi-level evaluation framework that balances signal-level audio quality with semantic, rhythmic, and structural video-music alignment. All metrics are computed using open-source libraries (NumPy [46], SciPy, Librosa [47], OpenCV [41], ImageBind [48]), ensuring transparency and reproducibility. This evaluation design is consistent with recent trends in video-to-music generation research and provides a solid quantitative foundation for the experimental analysis in CHAPTER 4: EXPERIMENT RESULTS.

CHAPTER 4: EXPERIMENT RESULTS

This chapter reports the experimental setup and evaluation results of the proposed system. It introduces the evaluation dataset, benchmarks the music understanding capability of the selected language model, and presents quantitative results for both music quality and video-music relationship metrics. Additionally, the chapter analyzes computational cost and compares the proposed approach with baseline methods to demonstrate its effectiveness and efficiency.

4.1 Evaluation Dataset

All experiments are conducted on a subset of the SymMV dataset [49], a publicly available benchmark consisting of music videos that exhibit strong semantic and temporal correspondence between visual content and background music. SymMV is particularly suitable for video-to-music evaluation because it contains professionally edited videos where musical structure, emotional progression, and visual narrative are closely aligned.

From the full dataset, 20 music videos are randomly selected to form the evaluation set. These videos span diverse visual themes, including urban scenes, sports activities, cinematic storytelling, and emotional narratives, ensuring that the evaluation reflects a wide range of semantic and affective contexts. The average video duration in the evaluation set is approximately 198.54 seconds, which allows assessment of long-range temporal coherence and scene-level musical consistency.

To ensure that the evaluation focuses exclusively on background music generation, all original audio tracks are preprocessed using the Demucs source separation library [50] to remove vocal components. This step eliminates potential interference from lyrics and singing voices, which are not the primary target of this study, and ensures that both ground-truth references and generated outputs are evaluated under consistent conditions.

4.2 Qwen3 on music theory benchmark

To justify the choice of a small language model for the music feature suggestion stage, we compared ChatMusician [20] and Qwen3 [45] variants on a MusicTheoryBench [20] and MMLU [51] for language understanding. This evaluation aims to assess each model’s ability to understand musical knowledge, perform reasoning, and generalize under both zero-shot and few-shot settings.

Models	MMLU	MusicTheoryBench			
		Zero-Shot Knowledge	Zero-Shot Reasoning	Few-Shot Knowledge	Few-Shot Reasoning
ChatMusician	41.58	41.26	22.45	41.64	26.53
ChatMusician-Base	43.54	41.26	25.51	42.75	26.53
Qwen3 1.7B	53.99	56.88	26.53	54.64	31.63
Qwen3 4B	66.81	**58.36**	28.57	56.13	33.67
Qwen3 4B 2507 Instruct	**68.87**	**58.36**	**32.65**	**56.88**	**34.69**

Table 2: Comparation between ChatMusician variant and Qwen3 variant in music theory understanding

The experimental results in Table 2 demonstrate that Qwen3-4B-Instruct-2507 [45] provides the strongest overall capability for music feature suggestion among the evaluated models. With the highest MMLU [51] score (68.87), the model shows a comprehensive understanding of music-related knowledge, which is essential for accurately identifying high-level attributes such as genre, instrumentation, and stylistic conventions from video captions. Compared to domain-specific baselines such as ChatMusician [20], the proposed model exhibits a substantial performance margin, indicating better generalization beyond narrowly defined music theory tasks.

More importantly, the advantage of Qwen3-4B-Instruct-2507 [45] becomes more pronounced in reasoning-oriented settings, which are critical for inferring abstract musical attributes like mood, style, and narrative context. The model achieves the best performance in both Zero-Shot Reasoning (32.65) and Few-Shot Reasoning (34.69), significantly outperforming smaller models and non-instruct variants. This suggests that instruction tuning enhances the model's ability to interpret implicit semantic cues and translate them into musically meaningful descriptors, rather than relying solely on factual recall.

These findings directly support the design choice of using Qwen3-4B-Instruct-2507 [45] as the semantic intermediary in the music feature suggestion stage. Strong zero-shot performance ensures robustness when processing diverse and unseen video content, while improved reasoning ability contributes to more consistent and interpretable feature extraction. As a result, the downstream prompt preparation and music generation stages benefit from higher-quality semantic guidance, leading to improved alignment between video content and generated music.

4.3 Music quality

Regarding music quality, the GroundTruth audio unsurprisingly exhibits strong harmonic stability, with a Total Harmonic Distortion (THD) value of 0.1603. This reflects the professional mixing and mastering processes typically applied to human-composed soundtracks. GVMGen [9] further reduces harmonic distortion, achieving the5 lowest THD value of 0.1339. In comparison, the proposed system yields a higher THD value of 0.2390. While this indicates increased harmonic richness, it does not necessarily imply inferior perceptual quality. MusicGen [10] generates audio directly in the waveform domain without explicit harmonic constraints, and the use of continuation-based generation across multiple segments may introduce mild harmonic accumulation. In the context of background music generation, such harmonic richness is acceptable and often characteristic of electronic and cinematic sound design.

Models	THD ↓	SF ↓	SCV ↑	ZCR ↓
GroundTruth	0.1603	0.0167	**718.7996**	0.0648
GVMGen [9]	**0.1339**	**0.0135**	709.33842	**0.0639**
Proposed pipeline	0.2390	0.0144	564.8818	0.0693

Table 3: Music quality evaluation results

Spectral Flatness (SF) values further clarify this observation. All evaluated models maintain very low SF values, indicating that none of the generated outputs degenerate into noise-like signals. The GroundTruth music records an SF value of 0.0167, while GVMGen [9] achieves 0.0135. Notably, the proposed system attains an SF value of 0.0144, which is lower than that of the GroundTruth. This confirms that despite its higher THD, the generated music remains predominantly tonal and harmonically structured rather than broadband or noisy.

Spectral Centroid Variability (SCV) reveals more distinct differences in temporal timbral behavior. The GroundTruth music exhibits a high SCV value of 718.80, indicating frequent changes in perceived brightness over time. This is expected in professionally composed soundtracks, which often involve instrument changes, dynamic orchestration, and expressive mixing. GVMGen [9] shows a similar level of variability with an SCV of 709.34. In contrast, the proposed system produces a significantly lower SCV of 564.88, suggesting more stable timbral characteristics throughout the video. This reduced variability aligns with the intended role of the generated audio as background music, where excessive timbral fluctuation could distract from visual content. Therefore, a lower SCV in this context reflects a design choice favoring perceptual stability over expressive complexity.

Zero-Crossing Rate (ZCR) results support this interpretation. The GroundTruth and GVMGen [9] outputs yield ZCR values of 0.0648 and 0.0639, respectively, while the proposed system records a slightly higher value of 0.0693. This indicates a marginal increase in high-frequency activity, which is commonly observed in electronic or ambient textures and does not signify excessive noise or instability.

4.4 Video-Music relationship

Beyond intrinsic audio quality, the video-music relationship metrics highlight the strengths of the proposed language-mediated pipeline. Cross-Modal Embedding Correlation (CMEC), which measures global semantic alignment between video and music using ImageBind embeddings, shows that the GroundTruth achieves the highest score of 0.2979. GVMGen [9] performs substantially worse with a CMEC of 0.1469, indicating limited semantic correspondence. In contrast, the proposed system achieves a CMEC score of 0.2456, significantly outperforming GVMGen [9] and approaching GroundTruth performance. This result demonstrates that the use of natural language as an intermediate representation effectively preserves high-level semantic information from video content through to the music generation stage.

Models	CMEC ↑	BMS ↑	CMSA ↑	SSM ↑
GroundTruth	**0.2979**	**0.2436**	0.1696	**0.2740**
GVMGen	0.1469	0.0913	0.1430	0.1188
Proposed pipeline	0.2456	0.0271	**0.1816**	0.1313

Table 4: Video-Music relationship evaluation results

Beat-Motion Synchrony (BMS) results reveal a different trend. The GroundTruth music exhibits strong rhythmic synchronization with visual motion, achieving a BMS score of 0.2436, reflecting deliberate manual alignment by human composers. GVMGen [9] achieves a moderate BMS score of 0.0913, benefiting from direct conditioning on visual motion features. The proposed system, however, records a lower BMS score of 0.0271. This outcome is expected, as the system is not explicitly conditioned on motion or beat-level visual cues. Instead, it prioritizes semantic and narrative coherence over precise rhythmic synchronization. In many cinematic and background music scenarios, music is intentionally designed to follow emotional progression rather than instantaneous motion, making lower BMS values acceptable within the intended application scope.

Temporal and structural alignment metrics further support the effectiveness of the proposed approach. Cross-Modal Similarity Alignment (CMSA), which evaluates the consistency of video-music similarity over time, yields a score of 0.1816 for the proposed system, outperforming both the

GroundTruth (0.1696) and GVMGen [9] (0.1430). This indicates that semantic correspondence between video and music is maintained uniformly across temporal segments. Finally, Self-Similarity Matrix (SSM) alignment reveals that the proposed system (0.1313) better preserves long-range structural correspondence with video content than GVMGen [9] (0.1188), although it remains below the GroundTruth reference (0.2740).

Overall, these results demonstrate that the proposed system achieves a favorable balance between audio stability and cross-modal semantic alignment. While it does not yet match diffusion-based models in beat-level synchronization or harmonic purity, it significantly outperforms them in semantic consistency and structural alignment with video content. This outcome directly supports the central hypothesis of this thesis: employing natural language as an intermediary modality enables more interpretable, controllable, and semantically aligned video-to-music generation, particularly suited for web-based and interactive applications.

4.5 Computational cost

All experiments were conducted on a Kaggle computing environment equipped with two NVIDIA Tesla T4 GPUs, each providing approximately 14.7 GB of VRAM and supporting CUDA with compute capability 7.5. The system also includes an x86_64 CPU with 4 processing cores and approximately 31.3 GB of system RAM. This hardware configuration enables efficient execution of large-scale deep learning models and supports GPU-accelerated inference and evaluation throughout the experimental pipeline.

Average values of test set	GVMGen	Proposed pipeline
Video duration	198.54	198.54
Computational time (seconds)	8197.75	**2245.19**
Memory usage (GB VRAM)	20.93	**10.57**
Number of parameters (Billions)	**2.2**	12.9

Table 5: Computational cost comparation between 2 systems

The computational cost comparison in Table *5* shows that the proposed pipeline is significantly more efficient than the GVMGen baseline in terms of runtime and memory consumption. Although both systems process videos of comparable average duration (198.54 seconds), the proposed method reduces the total computational time from 8197.75 seconds to 2245.19 seconds, corresponding to a speed-up of more than 3.6 times. In addition, GPU memory usage is reduced by approximately 50%, indicating that the modular design of

the proposed pipeline avoids the high memory overhead typically associated with end-to-end generative models.

Components of proposed system	Number of parameters (Billions)	Huggingface model names
Clustering	0.4	openai/clip-vit-large-patch14
Captioning	7	OpenGVLab/VideoChat-R1_5-7B
Music Feature suggestion and Prompt preparation	4	Qwen/Qwen3-4B-Instruct-2507
Music Generation	1.5	facebook/musicgen-medium

Table 6: Number of parameters from each components in proposed pipeline

Despite using a larger total number of parameters (12.9B vs. 2.2B), as detailed in Table *6*, the proposed pipeline achieves better computational efficiency by distributing parameters across specialized components that operate independently and sequentially in inference mode. In contrast, GVMGen relies on a monolithic architecture where most parameters are active simultaneously during generation, leading to higher memory usage and longer execution time. This result demonstrates that a modular, multi-stage pipeline can achieve superior computational efficiency even when incorporating multiple large pretrained models, making the system more practical for real-world video-to-music generation applications.

CHAPTER 5: CONCLUSION AND FUTURE WORK

This thesis presented a language-mediated framework for video-to-music generation, in which natural language is employed as an intermediate representation between visual understanding and music synthesis. Unlike conventional end-to-end multimodal approaches that directly map visual features to audio signals, the proposed system explicitly separates semantic reasoning from sound generation. This design choice improves interpretability, controllability, and modularity, making the system particularly suitable for web-based and interactive applications.

The proposed pipeline integrates multiple open-source components, including scene segmentation, video captioning, music feature inference, prompt preparation, and text-to-music generation, into a unified end-to-end system. By leveraging large vision-language models for semantic video understanding and small language models for music-theoretical reasoning, the framework enables high-level alignment between video content and generated background music while remaining computationally efficient. Experimental results on the SymMV dataset demonstrate that this modular design achieves a favorable balance between semantic coherence, audio stability, and practical deployability.

Quantitative evaluation shows that the proposed approach significantly outperforms the GVMGen baseline in terms of semantic alignment between video and music. In particular, the system achieves a Cross-Modal Embedding Correlation (CMEC) score of 0.2456, approaching the GroundTruth reference and substantially exceeding the baseline model. Furthermore, the highest Cross-Modal Similarity Alignment (CMSA) score among all evaluated methods indicates that semantic correspondence is consistently preserved across temporal segments. These results validate the core hypothesis of this thesis: natural language serves as an effective intermediary modality for preserving high-level semantics in video-to-music generation.

From an audio quality perspective, the generated music exhibits stable spectral characteristics that are well suited for background accompaniment. Although the system shows higher harmonic distortion compared to diffusion-based baselines, its low spectral flatness and reduced spectral centroid variability indicate tonal stability and controlled timbral evolution. This trade-off reflects a deliberate design choice favoring perceptual consistency and narrative support over expressive complexity, which aligns with the intended role of background music in video applications. However, beat-level synchronization with visual motion remains limited, as the system prioritizes semantic and emotional coherence rather than explicit rhythmic conditioning.

Despite these promising results, several limitations highlight opportunities for future work. First, rhythmic alignment could be improved by incorporating motion-aware conditioning into the music generation process. Visual motion cues such as optical flow magnitude, scene dynamics, or detected visual beats could be integrated as auxiliary signals alongside semantic prompts. This hybrid semantic-motion conditioning strategy has the potential to enhance Beat-Motion Synchrony while preserving the strengths of the language-mediated design.

Second, the current system relies on a single static prompt per scene cluster. Future extensions could explore adaptive or hierarchical prompting, where prompts evolve dynamically within a scene based on changes in visual content or inferred narrative progression. Such an approach may improve long-range temporal alignment and structural correspondence, particularly in complex or long-form videos, and could positively impact Self-Similarity Matrix (SSM) alignment metrics.

Third, while this thesis exclusively employs pre-trained open-source models, future research may investigate domain-specific fine-tuning on paired video-music datasets. Fine-tuning the music generation model on language-conditioned video data could strengthen the association between semantic descriptors and musical realizations, reduce harmonic artifacts, and improve overall audio quality. Lightweight adaptation techniques such as Low-Rank Adaptation (LoRA) would allow such improvements without incurring prohibitive computational costs.

In addition to objective evaluation, incorporating human perceptual studies represents an important future direction. Subjective listening tests focusing on emotional appropriateness, immersion, and narrative coherence would provide valuable complementary insights beyond embedding-based metrics, helping to assess whether quantitative improvements translate into perceptual gains for end users.

Finally, the web application itself can be extended to support interactive, human-in-the-loop music creation. Allowing users to modify inferred music features, regenerate specific segments, compare alternative styles, or preview results in real time would transform the system from a one-shot generator into a creative assistant. Such interactivity aligns with the broader vision of generative AI as a collaborative tool rather than a fully automated replacement.

In conclusion, this thesis establishes a solid foundation for language-mediated video-to-music generation, demonstrating that natural language can effectively unify vision understanding, music reasoning, and user intent within a single open-source framework. By combining semantic interpretability, computational efficiency, and modular design, the proposed system advances

the state of the art toward more controllable, accessible, and interactive multimodal music generation.

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APPENDICES